The Zinin reaction is a method used to convert aromatic nitro substituents (R–NO_2) into aromatic primary amines (R–NH_2) via the addition of a sulfide compound under basic conditions. In the process, some S^{2-} ions are also converted into $S_2O_3^{2-}$ ions, which have sulfur atoms with an oxidation number of +2.

During an experiment, a chemist performed the Zinin reaction with 2,7-dinitrofluorenone to produce 2,7-diaminofluorenone, as summarized in Figure 1.

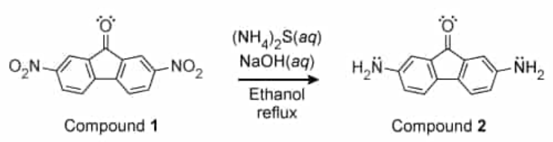


Figure 1 Conversion of 2,7-dinitrofluorenone (Compound **1**) to 2,7-diaminofluorenone (Compound **2**) via the Zinin reaction (Note: The molar masses of Compounds **1** and **2** are 270.20 g/mol and 210.24 g/mol, respectively.)

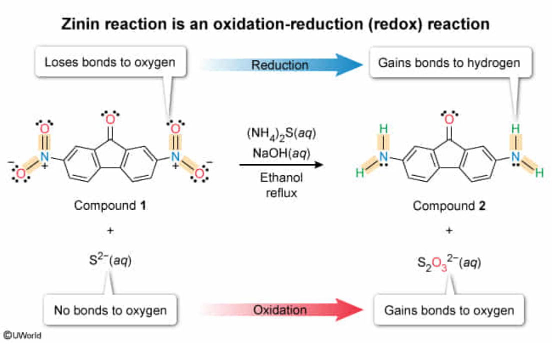
The conversion shown in Figure 1 was accomplished by adding 0.025 mol $(NH_4)_2S$(*aq*) and 0.025 mol NaOH(*aq*) into a stirred, yellow slurry containing 1.35 g of Compound **1** in 65 mL of ethanol. The mixture was continuously stirred and heated at reflux for 8 hours. Filtration, purification, and drying of the resulting dark purple precipitate yielded Compound **2** as cottony, dark purple needles.

The Zinin reaction of Compound **1** described in the passage is a:

- A. single replacement reaction. [7%]
- B. decomposition reaction. [30%]
- C. combination reaction. [6%]
- ✔ D. redox reaction. [56%]

Correct
56% answered correctly

Explanation:



Chemical reactions can often be **classified** according to common reaction patterns, but they can sometimes be more conveniently classified based on the results of the reaction. For example, in an oxidation-reduction (redox) reaction, the oxidation states of some atoms change during the conversion of the reactants to the products (ie, oxidation and reduction occur).

The occurrence of oxidation or reduction can often be identified without evaluating the oxidation numbers by simply examining the bonding partners of a given atom before and after a reaction. Generally, atoms that gain bonds to oxygen or lose bonds to H are being oxidized. Conversely, atoms undergoing the opposite process (ie, losing bonds to oxygen or gaining bonds to hydrogen) are usually being reduced.

In the given Zinin reaction of Compound **1**, the N atoms of the $–NO_2$ substituents lose bonds to O atoms and gain bonds to H atoms (ie, reduction). Simultaneously, the S^{2-} ion in $(NH_4)_2S$ is converted to the polyatomic ion $S_2O_3^{2-}$ by gaining bonds to O atoms (ie, oxidation). Therefore, the reaction is a redox reaction.

(Choice A) Single-replacement reactions involve the reaction of a compound with a free *element* (ie, a substance in its elemental form not bonded with other types of atoms) and results in the formation of a new compound and a new *element* as the products. The given Zinin reaction does not produce any free elements.

(Choice B) Decomposition reactions involve one compound breaking apart into two or more products. Compound **1** has some of its bonds changed but it does not break apart.

(Choice C) Combination reactions involve two or more species chemically combining and forming one product species. The given Zinin reaction forms more than one product.

Educational objective:
Chemical reactions can often be classified according to common reaction patterns, but they can also be further classified based on the results of the reaction. In oxidation-reduction (redox) reactions, a chemical conversion causes a change in the oxidation state of one or more of the constituent elements.

Time spent: 0 sec
Subject: General Chemistry

QID: 403447
System: Interactions of Chemical Substances

The Zinin reaction is a method used to convert aromatic nitro substituents (R–NO_2) into aromatic primary amines (R–NH_2) via the addition of a sulfide compound under basic conditions. In the process, some S^{2-} ions are also converted into $S_2O_3^{2-}$ ions, which have sulfur atoms with an oxidation number of +2.

During an experiment, a chemist performed the Zinin reaction with 2,7-dinitrofluorenone to produce 2,7-diaminofluorenone, as summarized in Figure 1.

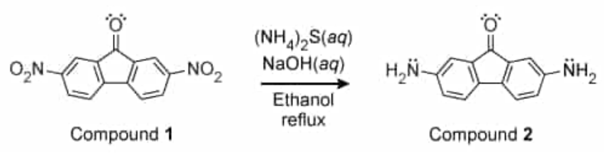


Figure 1 Conversion of 2,7-dinitrofluorenone (Compound **1**) to 2,7-diaminofluorenone (Compound **2**) via the Zinin reaction (Note: The molar masses of Compounds **1** and **2** are 270.20 g/mol and 210.24 g/mol, respectively.)

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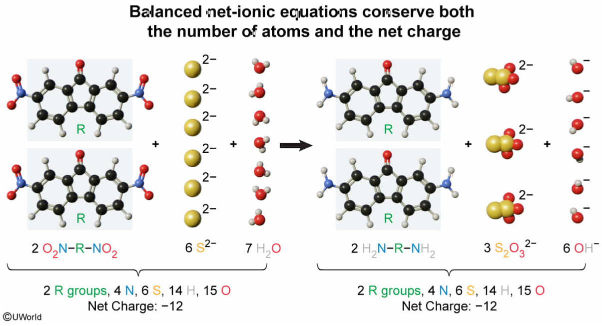
Which of the following gives the net ionic equation for the Zinin reaction described in the passage and is correctly balanced with respect to both the number of each type of atom and the net charge? (Note: R represents the three fused rings and C=O bond in the central structure of Compounds **1** and **2**, and the structure of the –NO_2 group gives its N atom an oxidation state of +3.)

- A. $O_2N–R–NO_2 + 6\ (NH_4)_2S + 5\ H_2O \rightarrow H_2N–R–NH_2 + 3\ (NH_4)_2S_2O_3 + 6\ NH_3 + 5\ H_2$ [7%]
- B. $O_2N–R–NO_2 + 2\ (NH_4)_2S + 5\ H_2O \rightarrow H_2N–R–NH_2 + (NH_4)_2S_2O_3 + 6\ OH^-$ [31%]
- C. $O_2N–R–NO_2 + 2\ S^{2-} + 5\ H_2O \rightarrow H_2N–R–NH_2 + S_2O_3^{2-} + 6\ OH^-$ [26%]
- D. $2\ O_2N–R–NO_2 + 6\ S^{2-} + 7\ H_2O \rightarrow 2\ H_2N–R–NH_2 + 3\ S_2O_3^{2-} + 6\ OH^-$ [34%]

Correct
34% answered correctly

Explanation:



Redox reactions are fully balanced when the same number of each type of atom and the same net charge are present on both sides of the reaction equation. As such, the correct answer to the question can be found most easily by counting the number of each type of atom and adding up the charges on each side of a reaction to identify and eliminate any of the given reaction choices that are unbalanced with respect to either the atoms or net charge. Of the given reactions, only Choice D is fully balanced in both respects.

Although much more difficult and time-consuming, the balanced equation can also be determined using half-reactions. Oxidation-reduction (redox) reactions consist of two simultaneous **half-reactions** (ie, one for the oxidation and one for the **reduction**) that add together to give the overall reaction. In a balanced redox reaction, the total number of electrons released from the oxidation half-reaction must equal the total number of electrons used by the reduction half-reaction. Consequently, the first step in balancing a redox reaction by half-reactions is to balance the number of electrons exchanged.

The structure of the –NO_2 group consists of an N=O bond, an $^{+}$N–O^{-} bond, and an N–R bond (ie, a total of four bonds to N), which causes the N atom to lack a pair of nonbonding electrons and gives its N atom an oxidation state of +3. In the reaction shown in Figure 1, both **N atoms are reduced** and go from an oxidation state of +3 in Compound **1** to −3 in Compound **2** (ie, a gain of 6 electrons per N atom).

Half-reaction 1 (reduction):2 N^{3+} + 12 e^- → 2 N^{3-}

Simultaneously, **S atoms are oxidized** and go from an oxidation state of −2 in the S^{2-} ions to +2 in the $S_2O_3^{2-}$ ions (ie, a loss of 4 electrons per S atom).

Half-reaction 2 (oxidation):S^{2-} → S^{2+} + 4 e^-

To balance and **cancel the electron terms** in the **overall reaction**, multiples of both half-reactions can be used and added together. For the multiples, a 2:6 ratio is used rather than a simpler 1:3 ratio to avoid a fractional coefficient in front of the $S_2O_3^{2-}$ when oxygen atoms are added back into the overall reaction.

(2 ×) Half-reaction 1: $4\ N^{3+} + \cancel{24\ e^-} \rightarrow 4\ N^{3+}$
(6 ×) Half-reaction 2: $6\ S^{2-} \rightarrow 6\ S^{2+} + \cancel{24\ e^-}$

Overall: $4\ N^{3+} + 6\ S^{2-} \rightarrow 4\ N^{3-} + 6\ S^{2+}$

Based on the electron balance above, the coefficients of the N and S species in the balanced reaction must be:

$$\mathbf{2}\ O_2\mathbf{N}\text{–R–}\mathbf{N}O_2 + \mathbf{6\ S^{2-}} \rightarrow \mathbf{2}\ H_2\mathbf{N}\text{–R–}\mathbf{N}H_2 + \mathbf{3\ S_2}O_3^{2-}$$

If a 1:3 ratio is used for the multiples in the previous half-reaction step, the reaction that can be multiplied by a factor of 2 to eliminate the fractional coefficient in front of $S_2O_3^{2-}$ and achieve the same result.

The next step is to **balance the atoms**. Because the reaction occurs in an **aqueous solution** under **basic conditions**, H_2O and OH^- can be added as reactants or products to complete balancing the equation. Therefore, the fully balanced net ionic equation for the given Zinin reaction is:

$$\mathbf{2\ O_2N\text{–R–}NO_2 + 6\ S^{2-} + 7\ H_2O \rightarrow 2\ H_2N\text{–R–}NH_2 + 3\ S_2O_3^{2-} + 6\ OH^-}$$

(Choice A) The atoms are not balanced in this reaction. The reactants have 58 H atoms, but the products have 56 H atoms.

(Choice B) The charges in this reaction are not balanced, and this is not a net ionic equation (ie, it contains NH_4^+ spectator ions).

(Choice C) The net charge is not balanced in this reaction. The net charge of the reactants is −4 but the net charge of the products is −8.

Educational objective:
Redox reactions are fully balanced when the same number of each type of atom and the same net charge are present on both sides of the reaction equation. For redox processes expressed as two half-reactions, the net reaction is found by adding the two half-reactions (or multiples of them) together so that the electron terms cancel.

Time spent: 0 sec
Subject: General Chemistry

QID: 403448
System: Interactions of Chemical Substances

The Zinin reaction is a method used to convert aromatic nitro substituents (R–NO_2) into aromatic primary amines (R–NH_2) via the addition of a sulfide compound under basic conditions. In the process, some S^{2-} ions are also converted into $S_2O_3^{2-}$ ions, which have sulfur atoms with an oxidation number of +2.

During an experiment, a chemist performed the Zinin reaction with 2,7-dinitrofluorenone to produce 2,7-diaminofluorenone, as summarized in Figure 1.

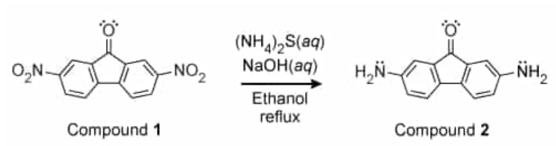


Figure 1 Conversion of 2,7-dinitrofluorenone (Compound **1**) to 2,7-diaminofluorenone (Compound **2**) via the Zinin reaction (Note: The molar masses of Compounds **1** and **2** are 270.20 g/mol and 210.24 g/mol, respectively.)

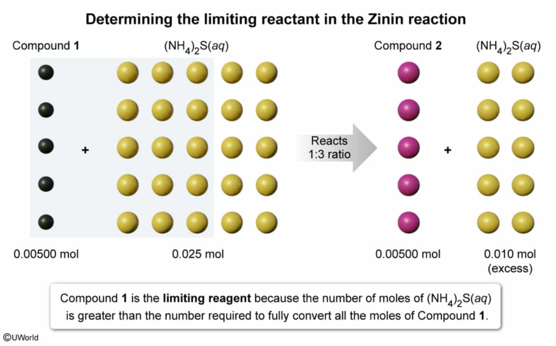
The conversion shown in Figure 1 was accomplished by adding 0.025 mol $(NH_4)_2S$(*aq*) and 0.025 mol NaOH(*aq*) into a stirred, yellow slurry containing 1.35 g of Compound **1** in 65 mL of ethanol. The mixture was continuously stirred and heated at reflux for 8 hours. Filtration, purification, and drying of the resulting dark purple precipitate yielded Compound **2** as cottony, dark purple needles.

If Compound **1** and $(NH_4)_2S$(*aq*) are found to react in a 1:3 molar ratio during the reaction shown in Figure 1, what is the limiting reactant in the reaction?

- ✔ A. Compound **1** [56%]
- B. $(NH_4)_2S$(*aq*) [33%]
- C. NaOH(*aq*) [4%]
- D. The limiting reactant cannot be determined without more information. [6%]

Correct
56% answered correctly

Explanation:



In a chemical reaction, the limiting reactant (ie, the reactant that gets used up first) determines the theoretical yield (ie, the maximum amount of product that can form) because the reaction stops once the limiting reactant is gone.

According to the passage, 1.35 g of Compound **1** is placed in the reaction vessel. Dividing this mass by the **molar mass** of Compound **1** gives the number of moles present in the vessel:

$$1.35\ \cancel{\text{g Compound } \mathbf{1}} \times \frac{1\ \text{mol Compound } \mathbf{1}}{270.20\ \cancel{\text{g Compound 1}}} = 0.00500\ \text{mol Compound } \mathbf{1}$$

If Compound **1** and $(NH_4)_2S(aq)$ are found to react in a 1:3 molar ratio, as stated in the question, the amount of $(NH_4)_2S(aq)$ needed to fully convert exactly 0.00500 moles of Compound **1** is:

$$0.00500\ \cancel{\text{mol Compound } \mathbf{1}} \times \frac{3\ \text{mol}\ (\text{NH}_4)_2\text{S(aq)}}{1\ \cancel{\text{mol Compound } \mathbf{1}}} = 0.0150\ \text{mol}\ (\text{NH}_4)_2\text{S(aq)}$$

The passage states that 0.025 mol of $(NH_4)_2S(aq)$ is added to the reaction vessel with Compound **1**. As such, the number of moles of $(NH_4)_2S(aq)$ is five times greater than the number of moles of Compound **1** and **exceeds the amount needed**.

$$0.025\ \text{mol}\ (\text{NH}_4)_2\text{S(aq) present} - 0.0150\ \text{mol}\ (\text{NH}_4)_2\text{S(aq) needed} = 0.010\ \text{mol}\ (\text{NH}_4)_2\text{S(aq) excess}$$

Therefore, Compound **1** is the **limiting reagent** because the number of moles of $(NH_4)_2S(aq)$ is greater than the number required to fully convert all the moles of Compound **1** present in the reaction mixture.

(Choice B) $(NH_4)_2S(aq)$ cannot be the limiting reactant because the number of moles of $(NH_4)_2S(aq)$ is in excess.

(Choice C) NaOH(*aq*) is present in molar excess to the same extent as $(NH_4)_2S(aq)$, but the amount is irrelevant because NaOH(*aq*) only provides the necessary basic conditions and does not participate directly in the reaction.

(Choice D) The given information allows a comparison of the number of moles of each species present in the reaction mixture relative to the reaction ratio. This information is sufficient to determine the limiting reactant.

Educational objective:

The limiting reactant is the reactant that gets used up first during a reaction. The limiting reactant determines the maximum amount of product that can form because the reaction stops once the limiting reactant is gone.

Time spent:0 sec	**QID**:403449
Subject:	**System:**
General Chemistry	Interactions of Chemical Substances

The Zinin reaction is a method used to convert aromatic nitro substituents (R–NO_2) into aromatic primary amines (R–NH_2) via the addition of a sulfide compound under basic conditions. In the process, some S^{2-} ions are also converted into $S_2O_3^{2-}$ ions, which have sulfur atoms with an oxidation number of +2.

During an experiment, a chemist performed the Zinin reaction with 2,7-dinitrofluorenone to produce 2,7-diaminofluorenone, as summarized in Figure 1.

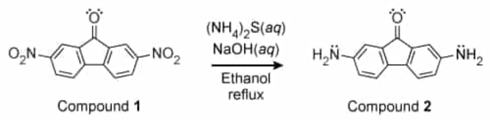


Figure 1 Conversion of 2,7-dinitrofluorenone (Compound **1**) to 2,7-diaminofluorenone (Compound **2**) via the Zinin reaction (Note: The molar masses of Compounds **1** and **2** are 270.20 g/mol and 210.24 g/mol, respectively.)

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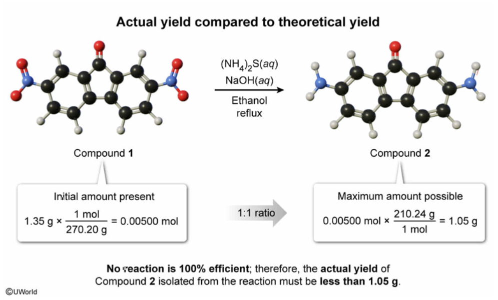
Based on the information in the passage, the mass of Compound **2** isolated from the reaction must be:

- A. greater than 1.05 g, because all the limiting reactant is converted into product. [16%]
- B. equal to 1.05 g, because conversion of the limiting reactant is perfectly efficient and there are no transfer losses during the experiment. [1%]
- C. less than 1.05 g, because conversion of the limiting reactant is not perfectly efficient and some small transfer losses occur during the experiment. [74%]
- D. equal to 1.35 g, because all the limiting reactant is converted into product.. [7%]

Correct
74% answered correctly

Explanation:



The theoretical yield of a reaction is the **maximum amount** of product that can form if all the limiting reactant is converted into products. According to the passage, 1.35 g of Compound **1** is placed in the reaction vessel. Dividing this mass by the molar mass of Compound **1** gives the number of moles present.

$$1.35\ \cancel{\text{g Compound } \mathbf{1}} \times \frac{1 \text{ mol Compound } \mathbf{1}}{270.20\ \cancel{\text{g Compound } \mathbf{1}}} = 0.00500 \text{ mol Compound } \mathbf{1}$$

Based on Figure 1, each molecule of Compound **2** that forms must be made from a molecule of

Compound **1** (ie, the reaction occurs in a **1:1 molar ratio**). If all the moles of Compound **1** are converted into Compound **2**, an equal number of moles of Compound **2** are formed; this is the maximum number of moles that can be produced in the reaction (ie, the theoretical yield).

$$0.00500\ \cancel{\text{mol Compound }\mathbf{1}} \times \frac{1\ \text{mol Compound }\mathbf{2}}{1\ \cancel{\text{mol Compound }\mathbf{1}}} = 0.00500\ \text{mol Compound }\mathbf{2}$$

Multiplying the theoretical yield of Compound **2** (in moles) by the molar mass of Compound **2** gives the **mass of the theoretical yield**.

$$0.00500\ \cancel{\text{mol Compound }\mathbf{2}} \times \frac{210.24\ \text{g Compound }\mathbf{2}}{1\ \cancel{\text{mol Compound }\mathbf{2}}} = 1.05\ \text{g Compound }\mathbf{2}\ (\text{theoretical yield})$$

However, **no reaction is 100% efficient**, and some losses in yield always occur (eg, incomplete reaction, transfer losses). Therefore, the mass of Compound **2** isolated from the reaction (ie, the **actual yield**) must be **less than 1.05 g**.

(Choice A) The amount of product that can be isolated from a reaction cannot exceed the theoretical yield. When all the moles of Compound **1** are consumed, no more of the Compound **2** product can form.

(Choice B) Isolating 100% of the theoretical yield is not achieved because no reaction is perfectly efficient, and some losses always occur.

(Choice D) The mass of Compound **2** that is isolated from the reaction cannot be equal to the initial mass of Compound **1** because the reaction occurs in a 1:1 molar ratio and the molar mass of Compound **2** is less than the molar mass of Compound **1**.

Educational objective:
The theoretical yield of a reaction is the maximum amount of product that can form if all the limiting reactant is converted into products. The actual yield from a reaction will always be less than the theoretical yield because no reaction is perfectly efficient.

Time spent: 0 sec
Subject: General Chemistry

QID: 403450
System: Interactions of Chemical Substances

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During an experiment, a chemist performed the Zinin reaction with 2,7-dinitrofluorenone to produce 2,7-diaminofluorenone, as summarized in Figure 1.

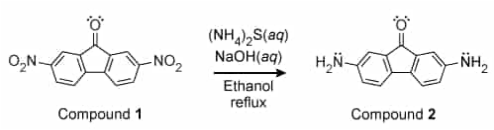

Figure 1 Conversion of 2,7-dinitrofluorenone (Compound **1**) to 2,7-diaminofluorenone (Compound **2**) via the Zinin reaction (Note: The molar masses of Compounds **1** and **2** are 270.20 g/mol and 210.24 g/mol, respectively.)

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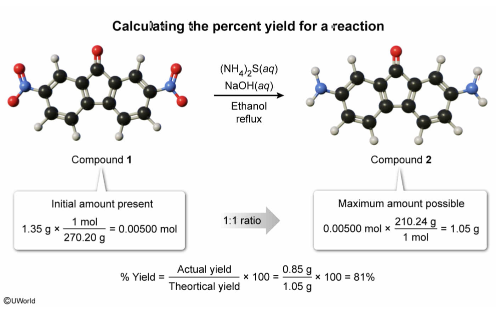
Suppose that 0.85 g of Compound **2** is isolated from the reaction described in the passage. What is the percent yield of Compound **2**?

- A. 63% [21%]
- B. 78% [18%]
- C. 81% [59%]
- D. 124% [1%]

Correct
59% answered correctly

Explanation:



The **percent yield** for a reaction assesses the extent to which the starting materials are converted into products during a reaction. As such, the percent yield for a given product is defined as:

$$\%\ \text{Yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

In the percent yield calculation, the theoretical yield is the **maximum amount** of the product that can form if all the limiting reactant is converted into products. As such, the theoretical yield is the basis against which the actual yield of the product is compared.

According to the passage, 1.35 g of Compound **1** is placed in the reaction vessel. Dividing this mass by the molar mass of Compound **1** gives the number of moles present.

$$1.35\ \cancel{\text{g Compound }\mathbf{1}} \times \frac{1\ \text{mol Compound }\mathbf{1}}{270.20\ \cancel{\text{g Compound }\mathbf{1}}} = 0.00500\ \text{mol Compound }\mathbf{1}$$

Based on the reaction shown in Figure 1, each molecule of Compound **2** that forms must be made from a molecule of Compound **1** (ie, the reaction occurs in a **1:1 molar ratio**). If all the moles of Compound **1** are converted into Compound **2**, an equal number of moles of Compound **2** are formed; this is the maximum number of moles that can be produced in the reaction (ie, the theoretical yield).

$$0.00500\ \cancel{\text{mol Compound }\mathbf{1}} \times \frac{1\ \text{mol Compound }\mathbf{2}}{1\ \cancel{\text{mol Compound }\mathbf{1}}} = 0.00500\ \text{mol Compound }\mathbf{2}$$

Multiplying the theoretical yield of Compound **2** (in moles) by the molar mass of Compound **2** gives the **mass of the theoretical yield**.

$$0.00500\ \cancel{\text{mol Compound }\mathbf{2}} \times \frac{210.24\ \text{g Compound }\mathbf{2}}{1\ \cancel{\text{mol Compound }\mathbf{2}}} = 1.05\ \text{g Compound }\mathbf{2}\ (\text{theoretical yield})$$

If 0.85 g of Compound **2** is isolated (ie, the actual yield from the reaction), the percent yield is:

$$\%\ \text{Yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100 = \frac{0.85\ \text{g}}{1.05\ \text{g}} \times 100 = 81\,\%$$

(Choice A) A value of 63% results from evaluating the wrong ratio:

$$\frac{\text{Actual yield}}{\text{Reactant mass}} \times 100 = \frac{0.85\ \text{g}}{1.35\ \text{g}} \times 100 = 63\,\%$$

(Choice B) A value of 78% results from evaluating the wrong ratio:

$$\frac{\text{Theoretical yield}}{\text{Reactant mass}} \times 100 = \frac{1.05\ \text{g}}{1.35\ \text{g}} \times 100 = 78\,\%$$

(Choice D) The percent yield can never exceed 100%, and valid values above 100% usually indicate contaminants. A value of 124% is invalid because it results from incorrectly evaluating the percent yield with an inverted ratio as:

$$\frac{\text{Theoretical yield}}{\text{Actual yield}} \times 100 = \frac{1.05\ \text{g}}{0.85\ \text{g}} \times 100 = 124\,\%$$

Educational objective:

The percent yield for a reaction is calculated as the ratio of the actual yield (the amount obtained) to the theoretical yield (the amount possible), expressed as a percentage. This value assesses the extent to which the starting materials are converted into products during a reaction.

Time spent: 0 sec
Subject: General Chemistry

QID: 403451
System: Interactions of Chemical Substances

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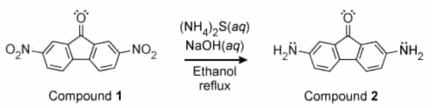

Figure 1 Conversion of 2,7-dinitrofluorenone (Compound **1**) to 2,7-diaminofluorenone (Compound **2**) via the Zinin reaction (Note: The molar masses of Compounds **1** and **2** are 270.20 g/mol and 210.24 g/mol, respectively.)

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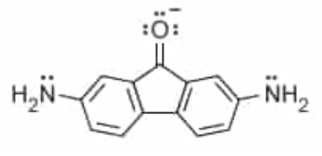
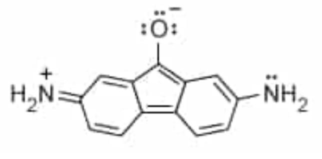
Which image shows a valid resonance structure of Compound **2**?

A.



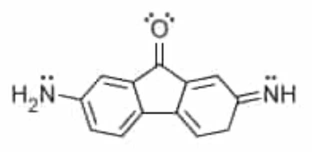
[4%]

B.



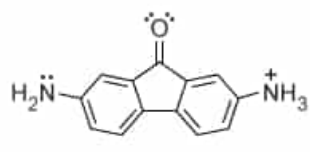
[80%]

C.



[7%]

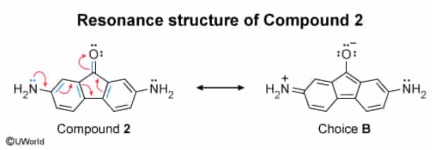
D.



[7%]

Correct
80% answered correctly

Explanation:



Some molecules cannot be fully represented using a single Lewis structure because some of the electrons can adopt more than one valid bonding configuration; this effect is known as **resonance**. In resonance, the **arrangement of the atoms** is **unchanged** but some π bonds and nonbonding electrons can "oscillate" from one atom to an adjacent atom in ways that do not violate the octet rule (apart from a valid exception). In effect, resonance causes the electrons to be **delocalized** (ie, "spread out") among the participating atoms.

Resonance can occur in conjugated systems with alternating single and multiple (ie, double or triple) bonds. For compounds that undergo resonance, a single Lewis structure represents only the bond configuration at a moment in time (ie, a resonance contributor). Because the electrons are in different places at different times, the true molecular structure is a resonance hybrid (ie, a weighted average of all resonance contributors).

In the structure of Compound **2**, the nonbonding electrons on the N atoms can migrate through the central aromatic rings by forming different π bond arrangements to reach the oxygen atom in the C=O bond. Therefore, structures representing valid bonding rearrangements caused by the migration of electrons (eg, Choice B) are valid resonance structures.

(Choice A) This structure has an extra pair of electrons that is not in the structure of Compound **2** (as seen in Figure 1). In resonance structures, the electrons move to different locations but the total number of electrons remains the same.

(Choice C) In this structure, one of the H atoms changes location (ie, from the N atom to the neighboring ring). Resonance structures involve *only* the movement of electrons.

(Choice D) This structure has an extra H atom that is not in the structure of Compound **2** (as seen in Figure 1). In resonance structures, the number, type, and arrangement of the atoms remains unchanged.

Educational objective:
Some molecules can adopt more than one valid bonding configuration by participating in resonance. In resonance, the arrangement of the atoms is unchanged but some π bonds and nonbonding electrons are delocalized among the participating atoms. In such cases, a single Lewis structure represents only a resonance contributor.

Time spent:0 sec
Subject:
General Chemistry

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System:
Interactions of Chemical Substances